

transition_model

December 10, 2025

```
[1]: %reload_ext autoreload
      %autoreload 2
```

```
[2]: # from kret_studies import *
      # from kret_studies.notebook import *
      # from kret_studies.complex import *

      # logger = get_notebook_logger()
```

```
[3]: from waymo_agent import *
      from waymo_agent.osmnx import *
      from waymo_agent.data_classes import *
      from waymo_agent.action_heuristic import *
      from waymo_agent.graph_env import *
      from waymo_agent.simulation import *
      from kret_sandbox.VIS import dtt
```

```
[4]: config = EnvConfig(max_new_requests_per_step=2)
```

```
[5]: """
      Define the observation space for the environment.
      - globals: day of week (sin, cos), time of day (sin, cos),
      ↪weather_one_hot(3) (7,) #TODO stretch goal: add demand per node, next step
      ↪speed limits
      - supply_demand_ratio: current supply-demand ratio, vehicle-per-node,
      ↪lambda-per-node (3,)
      - vehicles: (num_vehicles, 7) -> [loc_x_norm, loc_y_norm, battery,
      ↪status_one_hot(4)]
      - pending_requests: (max_pending_requests, 9) -> [pickup_x_norm,
      ↪pickup_y_norm, dropoff_x_norm, dropoff_y_norm, distance_meters, est_cost,
      ↪max_wait_time, wait_time, status_one_hot(1)] # TODO add max wait time before
      - active_rides: (num_vehicles, 8) -> [pickup_x_norm, pickup_y_norm,
      ↪dropoff_x_norm, dropoff_y_norm, price, total_trip_distance,
      ↪trip_distance_remaining, pickup_distance_remaining]
      - dispatch_mask: (num_vehicles) -> 1 if vehicle can be dispatched to
      ↪request, else 0
```

```

        - pricing_mask: (max_pending_requests) -> 1 if request needs pricing,
        ↪ decision, else 0
    """
    """

```

```
[5]: ''
```

```
[6]: # config = ActiveRideDF.space_config(config, 10)
# config["shape"]
```

```
[7]: env = RideShareEnv(config=config)
G = env.G
cfg = env.config
```

```

/Users/Akseldkw/micromamba/envs/kret_312/lib/python3.12/site-
packages/gymnasium/spaces/box.py:236: UserWarning: WARN: Box low's
precision lowered by casting to float32, current low.dtype=float64
    gym.logger.warn(
/Users/Akseldkw/micromamba/envs/kret_312/lib/python3.12/site-
packages/gymnasium/spaces/box.py:306: UserWarning: WARN: Box high's
precision lowered by casting to float32, current high.dtype=float64
    gym.logger.warn(
/Users/Akseldkw/micromamba/envs/kret_312/lib/python3.12/site-
packages/gymnasium/spaces/box.py:424: UserWarning: WARN: Casting input x to
numpy array.
    gym.logger.warn("Casting input x to numpy array.")

```

```
[8]: s, info = env.reset()
```

```

/Users/Akseldkw/micromamba/envs/kret_312/lib/python3.12/site-
packages/gymnasium/spaces/box.py:424: UserWarning: WARN: Casting input x to
numpy array.
    gym.logger.warn("Casting input x to numpy array.")

```

```
[9]: veh = env.observation_curr["vehicles"]
req = env.observation_curr["pending_requests"]
rides = env.observation_curr["active_rides"]
type(veh), type(req), type(rides)
```

```
[9]: (waymo_agent.data_classes.vehicles.VehicleDF,
waymo_agent.data_classes.requests.RequestDF,
waymo_agent.data_classes.active_rides.ActiveRideDF)
```

```
[10]: veh.shape, req.shape, rides.shape
```

```
[10]: ((2, 6), (25, 21), (2, 18))
```

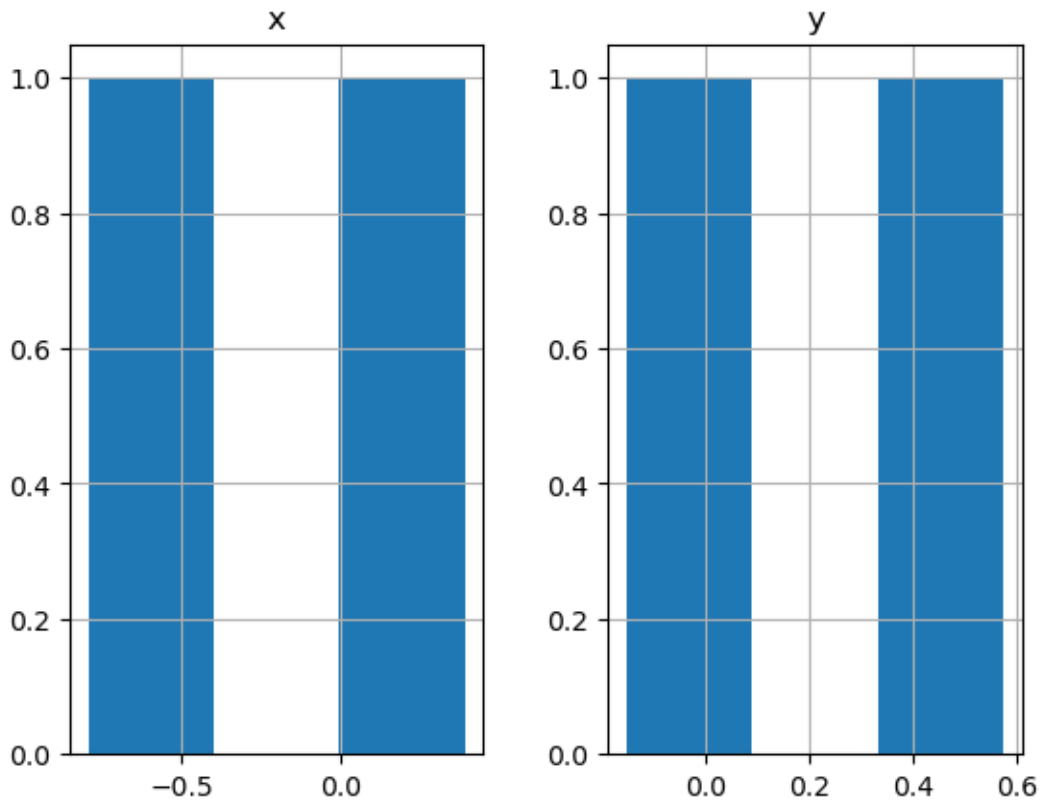
```
[11]: cfg = env.config  
      cfg.lambda_per_node * env.num_nodes
```

```
[11]: 0.10426666666666667
```

```
[12]: action = env.action_space.sample()  
      dispatches = pd.DataFrame({"dispatch": action["dispatch"]})  
      prices = pd.DataFrame({"prices": action["prices"]})  
      reposition = pd.DataFrame(action["reposition"], columns=["x", "y"])
```

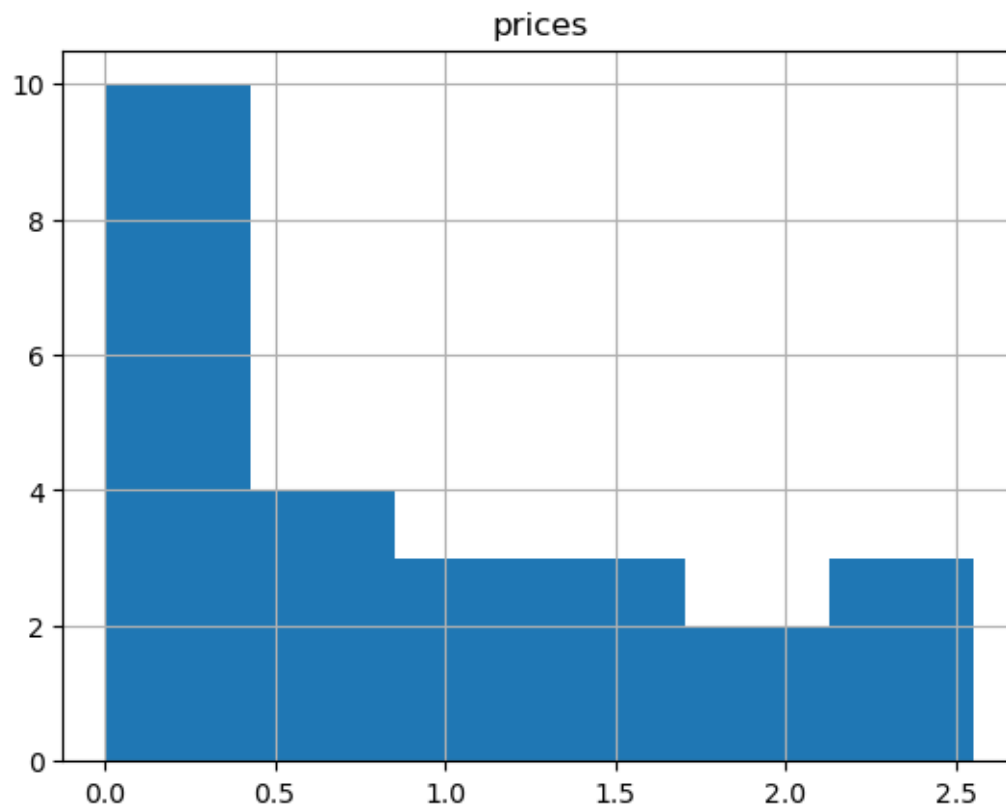
```
[13]: reposition.hist(bins="rice")
```

```
[13]: array([[<Axes: title={'center': 'x'}>, <Axes: title={'center': 'y'}>]],  
          dtype=object)
```



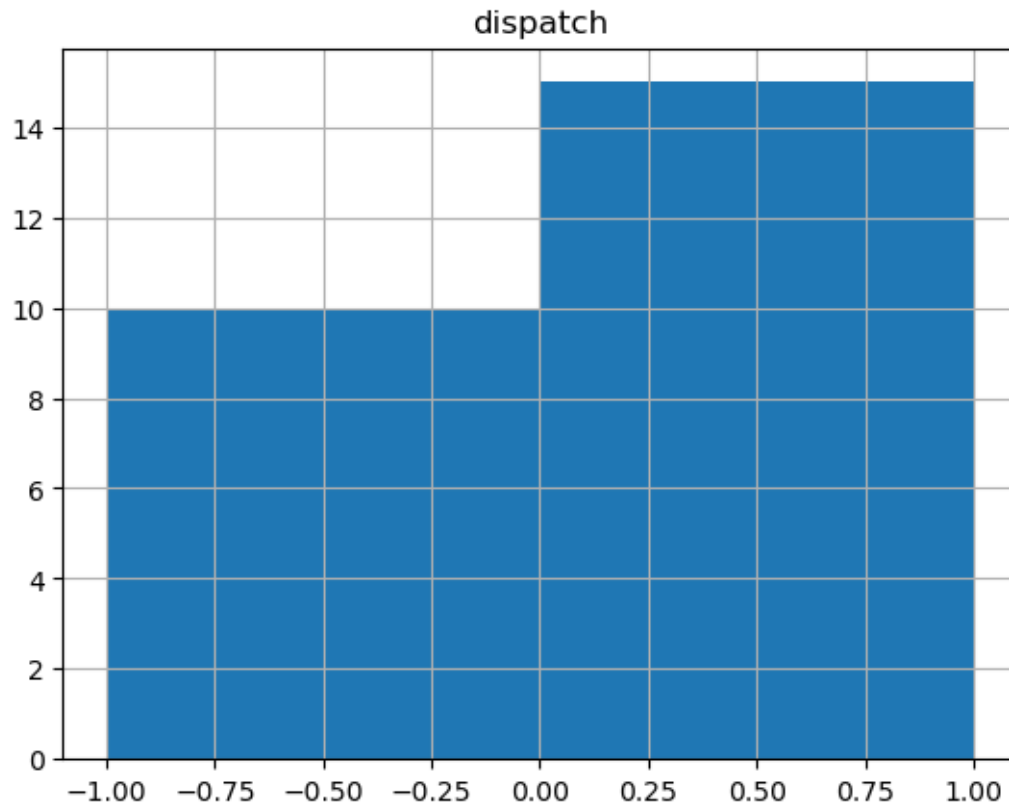
```
[14]: prices.hist(bins="rice")
```

```
[14]: array([[<Axes: title={'center': 'prices'}>]], dtype=object)
```



```
[15]: dispatches.hist(bins="rice")
```

```
[15]: array([[<Axes: title={'center': 'dispatch'}>]], dtype=object)
```



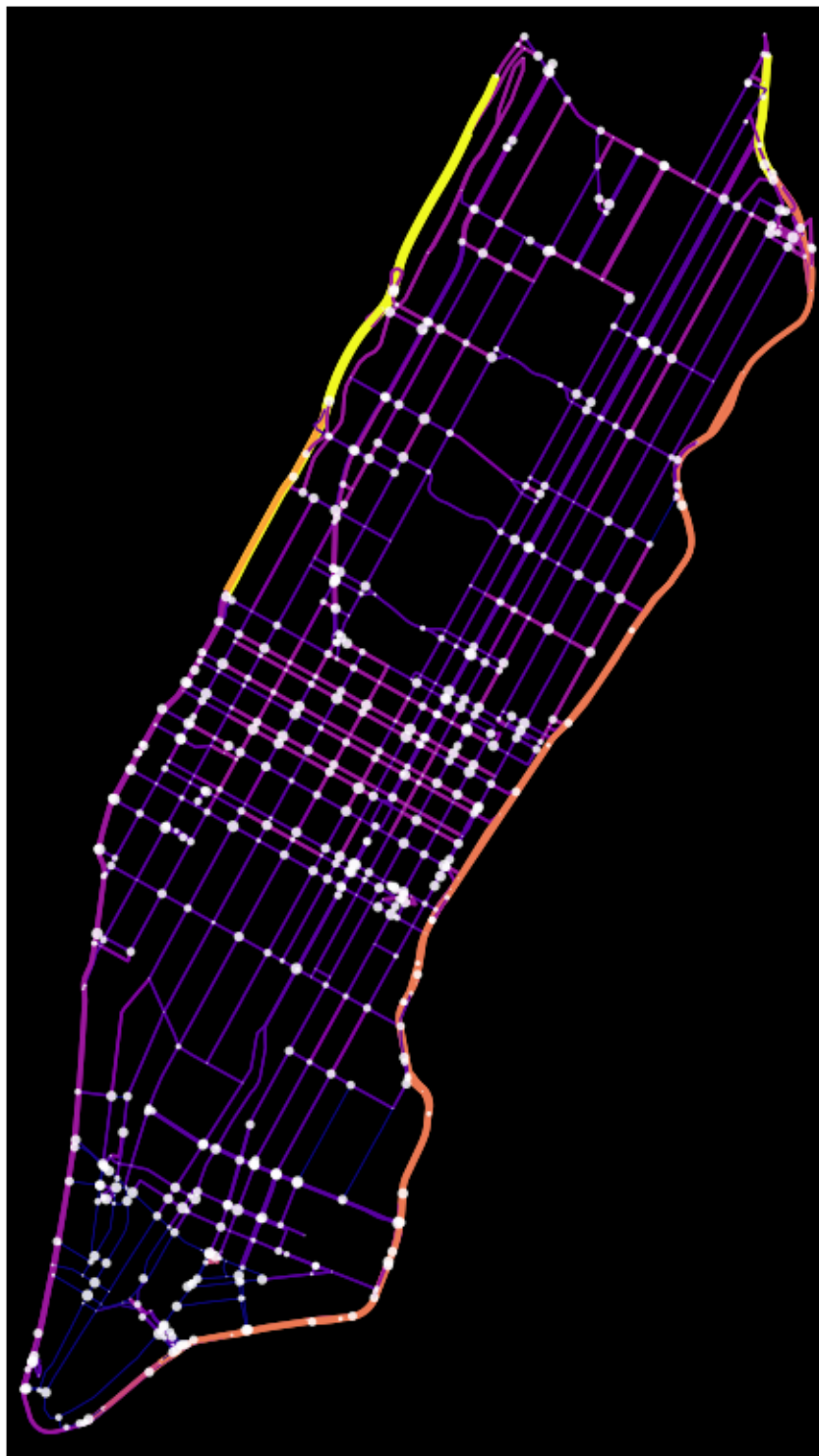
```
[24]: unembed_L2(G, veh.loc_x_norm, veh.loc_y_norm)
```

```
[24]: (array([584336.68564209, 589702.72233865]),
      array([4506714.06317052, 4518377.00151671]))
```

```
[26]: unembed_L2_env(env, veh.loc_x_norm, veh.loc_y_norm)
```

```
[26]: (7064.555262521739, 7179.2898836777495)
```

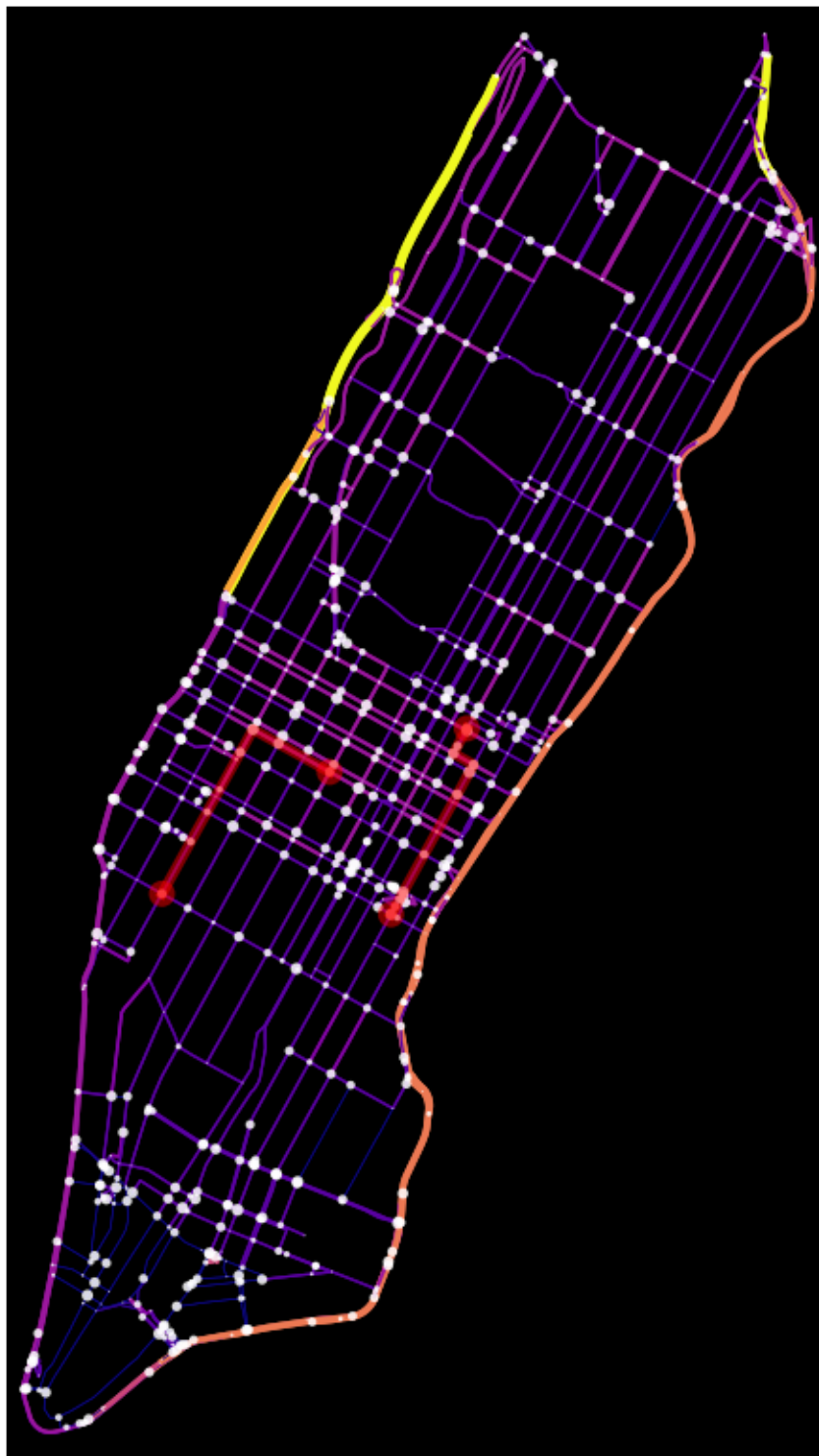
```
[20]: fig, ax = env.render()
      fig
```



Rendered step 0 with 0 active rides, 2 pending requests, and 2 vehicles.
Plot bounds: `x=(np.float64(-74.018673238), np.float64(-73.927302062))`,

```
y=(np.float64(40.6987979), np.float64(40.8212371))
Vehicle positions (projected): x=0    -1868.607311
1    3497.429373
Name: loc_x_norm, dtype: float64, y=0    -5304.579949
1    6358.358370
Name: loc_y_norm, dtype: float64
Vehicle positions (normalized): x_norm=[-0.2514  0.5003], y_norm=[-0.7488  0.885
]
```

[20]:



```
[21]: env.node_df.sample(4)
```



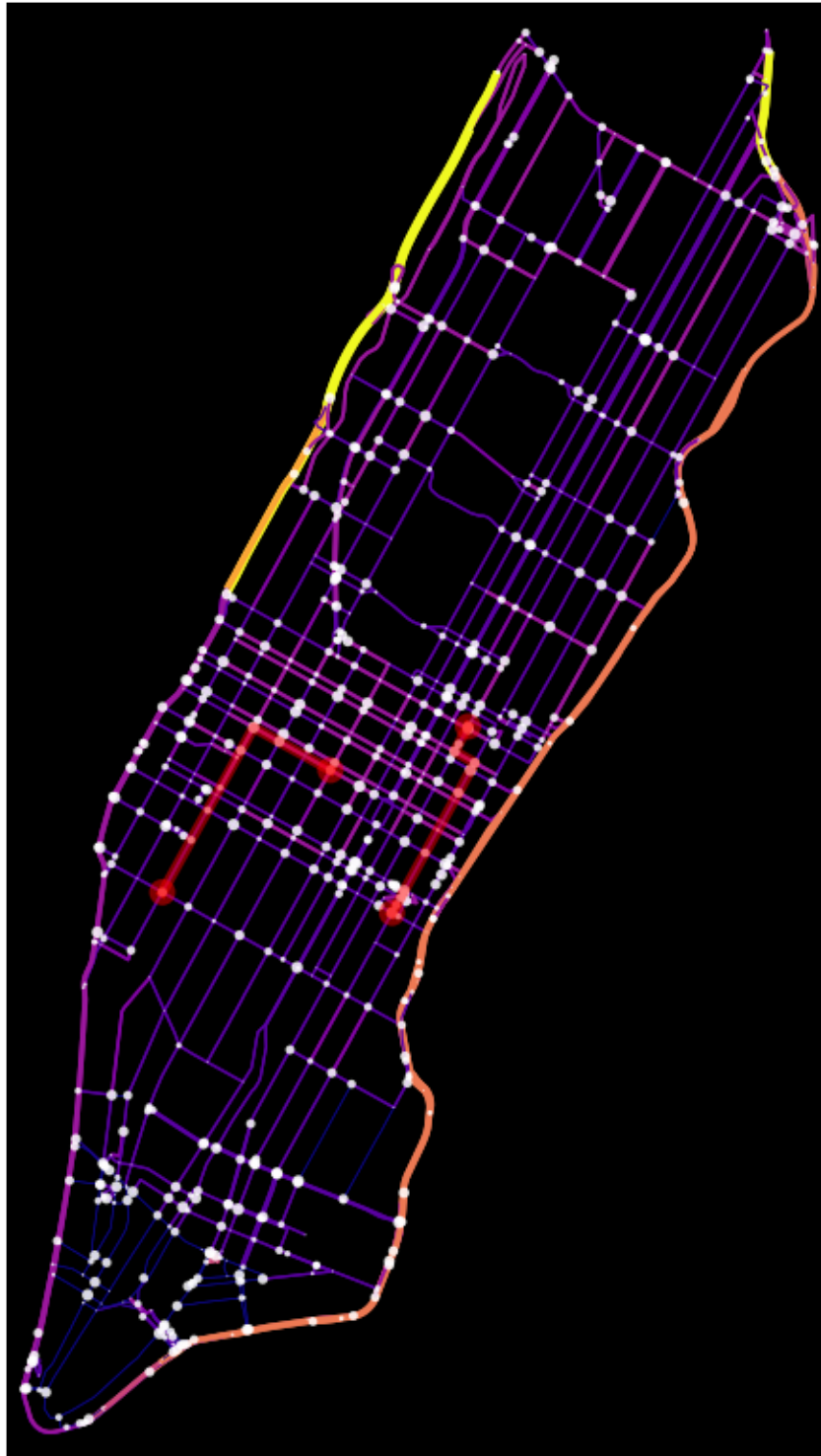
```
[21]:
```

	node_id	y	x	street_count	lambda	x_centered	\
81	42428588	40.772656	-73.982286	4	0.0084	-245.9404	
134	42430736	40.768370	-73.992786	4	0.0079	-1126.5670	
156	42432825	40.762295	-73.972365	4	0.0076	604.7833	
575	1701844698	40.747991	-73.970396	3	0.0138	789.6647	

	y_centered	x_norm	y_norm
81	1958.9608	-0.0345	0.2744
134	1472.9777	-0.1578	0.2063
156	818.5375	0.0847	0.1147
575	-767.3084	0.1106	-0.1075

```
[22]: ax.scatter(40.5, -73.5, c="red", s=100, zorder=20)
ax.scatter(40.7, -73.7, c="blue", s=100, zorder=20)
# ax.scatter()
fig
```

```
[22]:
```



[]:

```
[ ]: (array([586184.85233191, 584695.75394838]),
      array([4513996.0826386 , 4512068.67821121]))
```

```
[ ]:
```

```
[ ]:   vehicle_id  loc_x_norm  loc_y_norm  battery  status  ride_id
0         0      0.0075    0.2713  0.712997      0      -1
1         1     -0.2011    0.0013  0.609400      0      -1
```

```
[ ]:
```